

Towards a Compute-Efficient Physics-Informed Neural Network

From Beam-Vibration Benchmark to Railway Physics

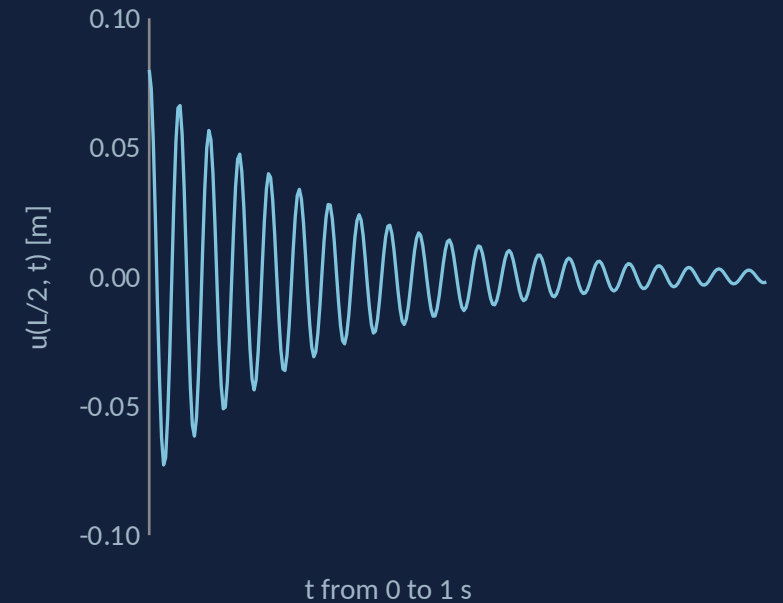
[Presenter name]

[Programme / institution]

Physics-informed neural networks · Euler–Bernoulli beam vibration · Railway engineering (target application)

Progress report · completed experiments through B2-E03 · PINN-RP commit cdd87b5

Analytical reference: mid-span displacement of the damped fixed–fixed beam (Mode 1)



20.58 Hz · 20.58 cycles in 1 s · decay 3.54 s^{-1}

The problem: accurate dynamics at low compute

Objective

Lower physical
error

+

Better dynamic
fidelity

+

Lower
computational cost

Tracked on every run



PDE residual



Frequency error



Damping / decay rate



Persistence (cycles)



Peak memory



Wall-clock time

A low scalar loss is not enough: $u \equiv 0$ also satisfies the homogeneous beam equation, so a collapsed late-time field can carry a small residual. Dynamics are therefore measured directly.

PDE
 $c^2 u_{xxxx} + u_{tt} + \gamma u_t = 0$

PINN
network $u_\theta(x, t)$

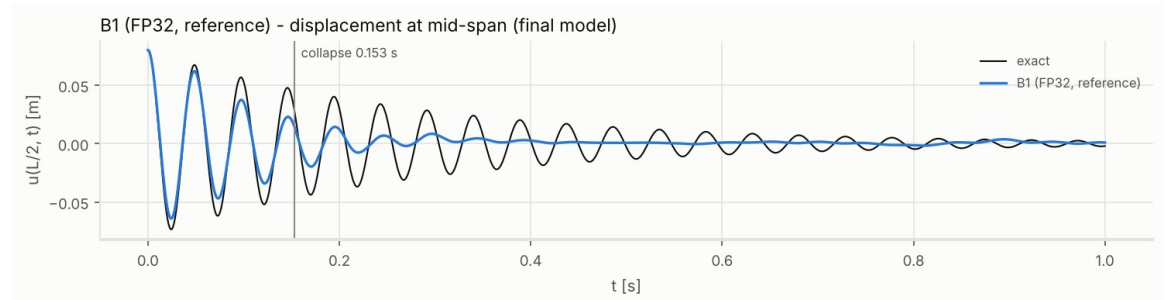
Prediction
vibration $u(x, t)$

Observed failure mode

Correct early oscillation

Artificial damping

Collapse



B1 (seed 1234): decay 8.98 s^{-1} vs exact 3.54 s^{-1} ; collapse at 0.153 s (3.15 of 20.58 cycles)

Starting point: the published benchmark

BENCHMARK

Cem Söyleyici & Hakkı Özgür Ünver
“A Physics-Informed Deep Neural Network based beam vibration framework for simulation and parameter identification”
Engineering Applications of Artificial Intelligence, 141 (2025) 109804 · DOI 10.1016/j.engappai.2024.109804

Method	Spatio-temporal Fourier feature mapping + NTK adaptive loss weighting
Network	6 × 200 tanh · Adam lr 1e-4 · 45,000 epochs
Benchmark FE-D-M1	Fixed-fixed damped beam · L = 2.75 m · Mode 1 · 20.58 Hz · 20.58 cycles in 1 s
Published accuracy	Relative L2 = 4.64e-4 (paper's reported value)

Our controlled re-implementation of the paper method (C0)

3.26 relative L2 · static (non-oscillating) field · NTK weights grew to $\sim 10^{14}$ – 10^{15}

640,000 PDE evaluations = 2.2 % of the paper's 28,800,000 (45,000 epochs × 640)

This is our incumbent, not our novelty.
Research question: can we reproduce it, diagnose why it loses dynamic fidelity, and improve physical accuracy per unit compute?

Research infrastructure achieved

ACHIEVED



Frozen ROOT provenance

Batch-1 repo frozen at d31864a; byte-identical after every phase



Independent PINN-RP

All new work in a separate repository; ROOT never written



Experiment registry

42 Batch-2 runs, each with config hash, git SHA, seed



Benchmark + exact reference

Closed-form damped Mode-1 solution; paper-faithful and exact-root references



Fixed evaluation

201×2001 grid; frozen persistence metric; 51×501 residual grid



Compute accounting

PDE evaluations, forward/backward passes, peak RAM, wall-clock



Pre-registration

Hypotheses and decision thresholds committed before any training



Seed replication

Seeds 1234–1236 for every Batch-2 arm; bit-exact control re-runs



Checkpoints + safety

Snapshots every 128k evaluations; runtime guard blocks writes to ROOT



Automated tests

168 tests passing at B2-E03 completion

What we tested

68

completed training runs

26 Batch 1 + 42 Batch 2

35

distinct configurations

25 Batch 1 + 10 new in Batch 2

16

intervention families

in 4 classes

3

seeds per Batch-2 arm

1234, 1235, 1236

Representation / formulation

- Fourier feature embedding (paper method)
- NTK adaptive loss weighting (paper method)
- Fourier convention / input scaling / bias init
- Hard IC/BC constraints
- $\tanh^2(\omega_1 t)$ temporal conditioning
- Mixed formulation $v = u_{xx}$
- Modal reduction (diagnostic)

Optimization / training

- Exponential LR decay
- Larger mini-batch (32 $\rightarrow$ 128)
- Training budget (5k $\rightarrow$ 20k steps)
- Adam $\rightarrow$ L-BFGS
- Full-batch Adam
- Causal temporal weighting

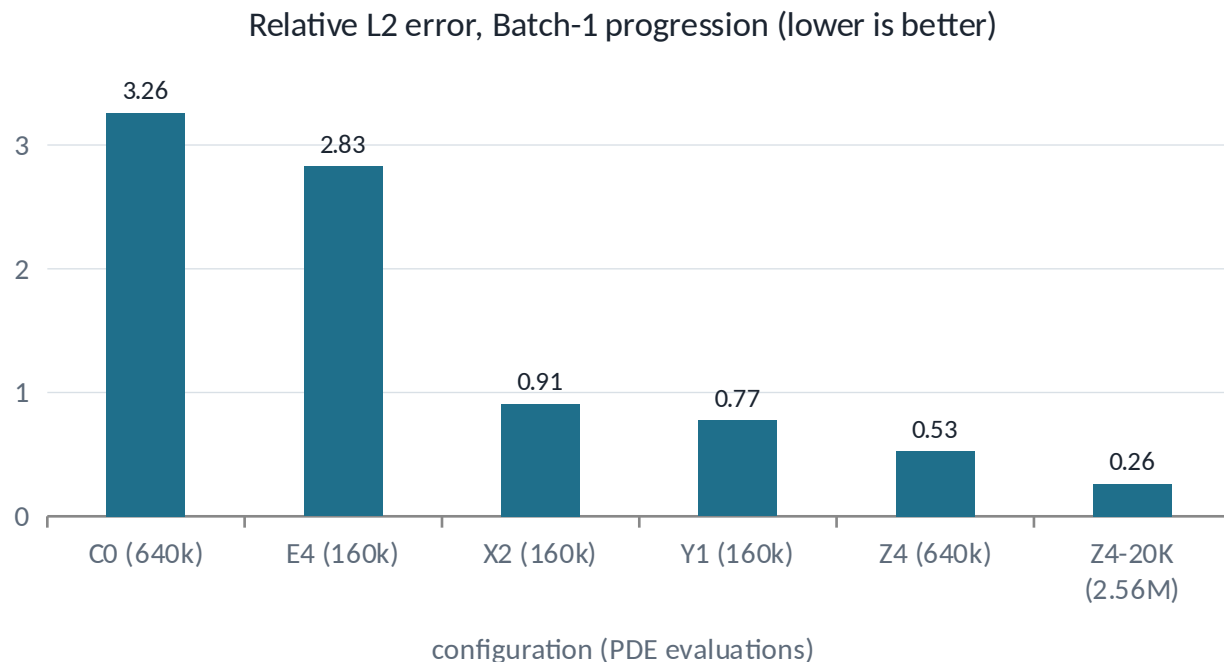
Sampling

- RAD adaptive sampling
- Collocation resampling / larger pool (L-BFGS)

Numerical precision

- FP64 (vs FP32)

Batch 1: reproduction and first optimizations



Persistence (cycles of 20.58): **Y1 1.17 → Z4 3.15 → Z4-20K 8.09** · C0, E4, X2: no sustained oscillation

Budgets differ (labels): E4/X2/Y1 at 160k, C0/Z4 at 640k, Z4-20K at 2.56M evaluations. Z4 = the frozen B1 reference used in Batch 2.

FAILED ON THIS BENCHMARK

Paper method (Fourier + NTK)

Static field under the tested budget (C0 L2 3.26)

ACHIEVED

Hard IC/BC constraints + $\tanh^2(\omega_1 t)$

IC/BC exact by construction ($4.8e-6$ / $1.4e-5$); required network output $O(1)$ instead of $\approx -8,369$

ACHIEVED

LR decay · mini-batch 128 · budget

Persistence 1.17 → 3.15 → 8.09 cycles

ACHIEVED

Supervised capacity check

Network recovers $\omega = 129.25 / 129.37$ rad/s (exact 129.32): capacity is not the bottleneck

FAILED ON THIS BENCHMARK

RAD adaptive sampling

Did not solve this benchmark failure under the tested budget

B2-E01: one change at a time, matched compute

Metric	B1	B1 FP64	Mixed v = u_xx	Modal q(t)
Persistence [cycles]	3.15	3.15	1.67	7.62
L2 (exact)	0.526	0.526	0.675	0.270
Frequency error	-0.96 %	-0.96 %	-2.0 %	-0.04 %
Decay rate [s^{-1}] (exact 3.54)	8.98	8.98	13.24	5.40
u PDE residual	0.110	0.110	2.57	0.074
Training time [s]	822	1,404	374	190
Peak RAM [MB]	934	1,178	730	704

Seed 1234 · every arm 640,000 PDE evaluations · 4 arms run concurrently on 4 vCPU

FAILED ON THIS BENCHMARK

FP64 precision

No accuracy change; 1.71× wall-clock, +26 % RAM

FAILED ON THIS BENCHMARK

Mixed formulation

0.46× wall-clock, but worse physics: u-residual 2.57 vs 0.110

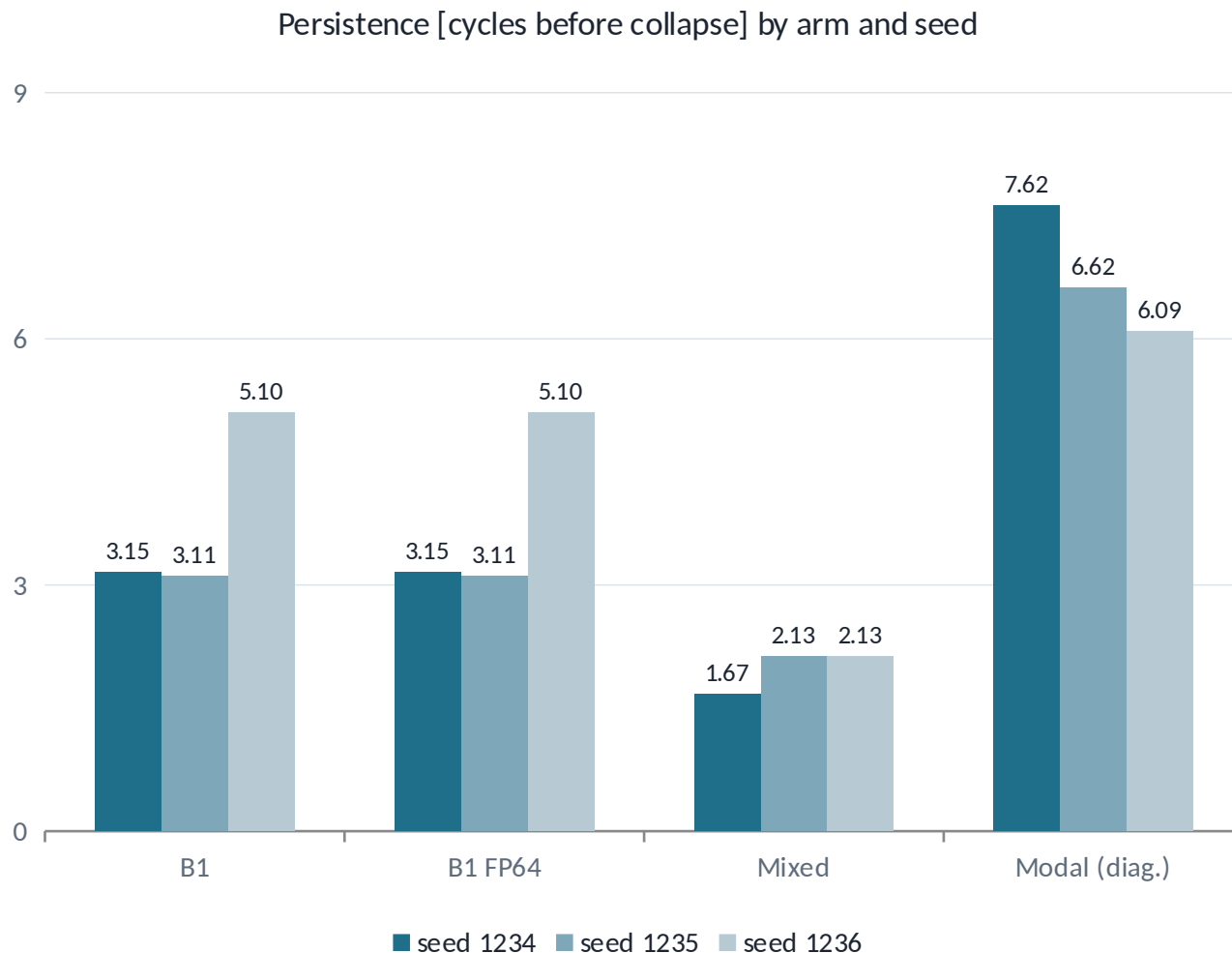
DIAGNOSTIC

Modal reduction

7.62 cycles, L2 0.270 — exact spatial mode supplied

*Removing the spatial learning burden
substantially improves temporal learning.*

Three-seed replication: what is robust



FAILED ON THIS BENCHMARK

FP64 $\equiv$ FP32 in 3/3 seeds

max relative L2 difference 5.4e-6; 1.62–1.71 $\times$ wall-clock

FAILED ON THIS BENCHMARK

Mixed: cheaper, never more accurate

0.46–0.47 $\times$ wall-clock; L2 0.623–0.675 vs B1 0.402–0.549

DIAGNOSTIC

Modal: longer persistence in 3/3 seeds

$\times 2.42$, $\times 2.13$, $\times 1.19$ vs B1; collapses at 0.296–0.370 s, not 1 s

DIAGNOSTIC

Modal improves, then degrades

peaks at 8.16–8.62 cycles (384k–512k), then retreats by 640k

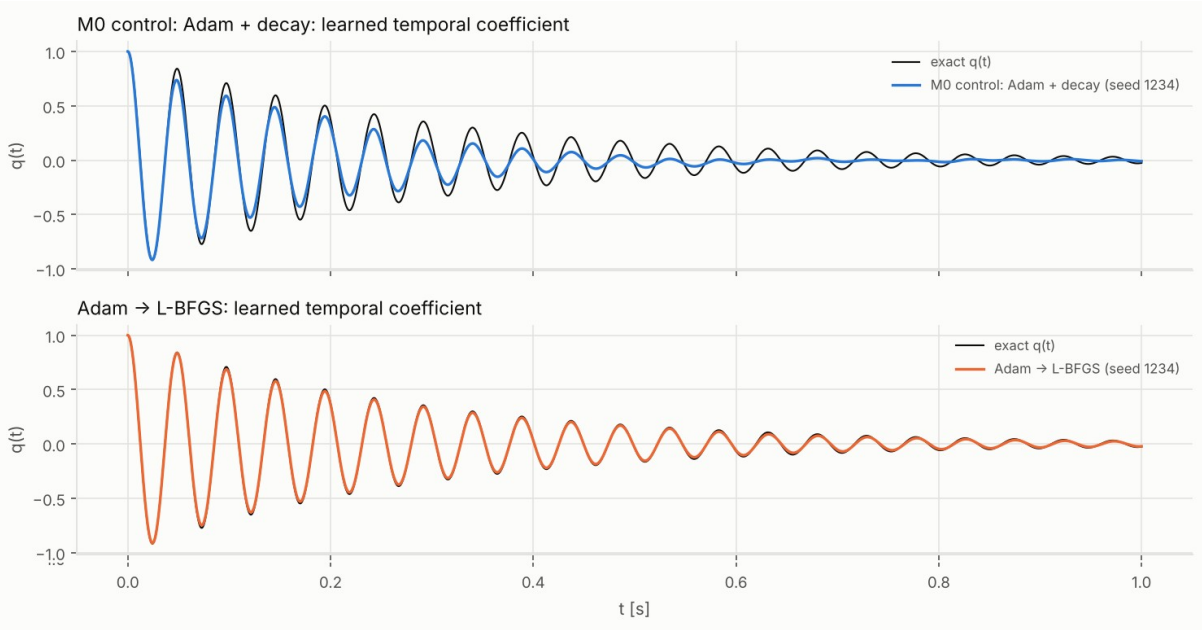
BENCHMARK

B1 seed variability

3.11–5.10 cycles; L2 0.402–0.549

B2-E02: the optimizer test (modal diagnostic)

DIAGNOSTIC



Arm (3 seeds)	Persistence [cycles]	L2	Decay [s ⁻¹]	Rel. wall-clock
M0 Adam + decay	6.1–7.6	0.27–0.33	5.4–6.1	1.00
Adam → L-BFGS	20.58	0.038–0.093	3.62–4.10	0.91–0.93
Causal weighting	2.1–4.6	0.56–1.56	fit not meaningful	1.00–1.01

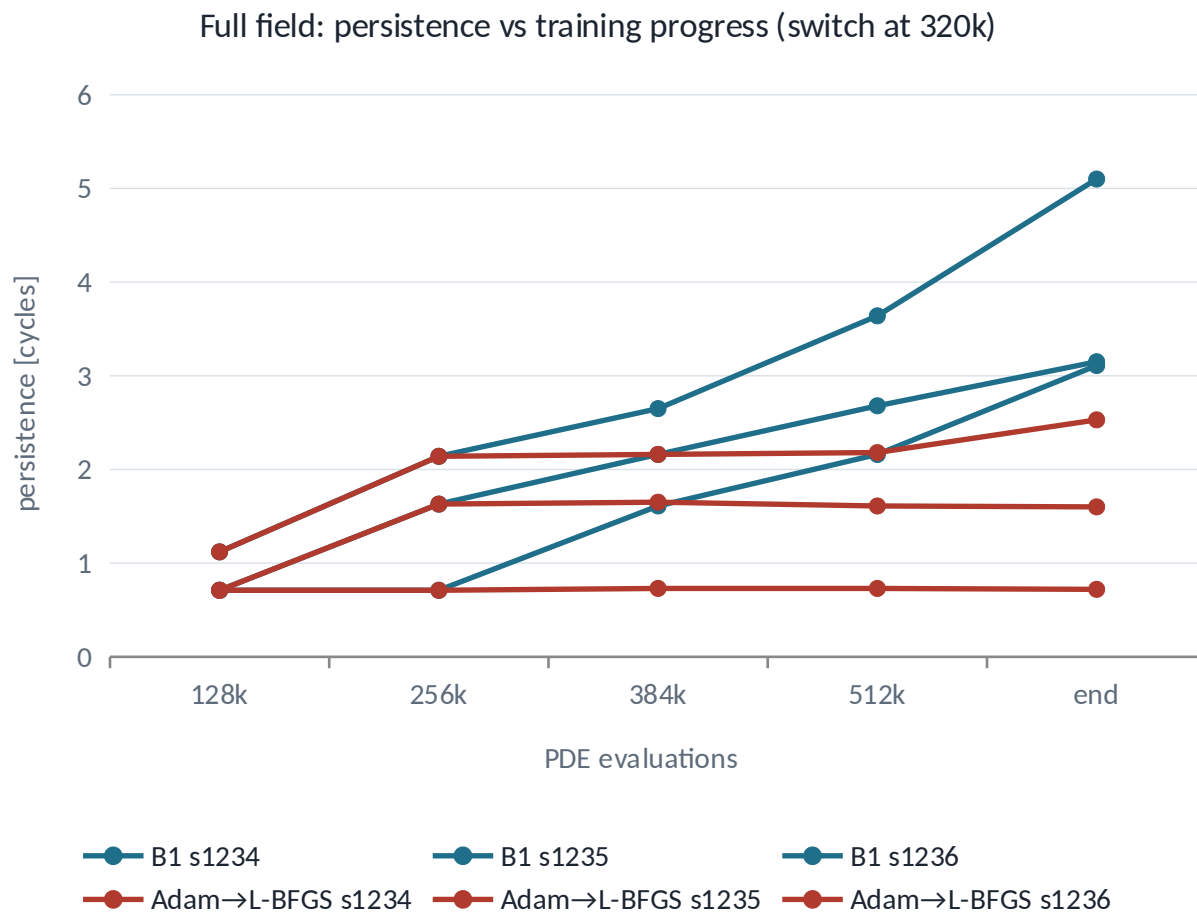
20.58
cycles = full 1-s window, 3/3 seeds

0.27–0.33 → 0.038–0.093
relative L2: Adam control → Adam → L-BFGS

Diagnostic only — not yet a valid full-field solution.
Exact spatial mode supplied · 623,360 vs 640,000 PDE evaluations · +29–30 % peak RAM

Transfer to the full field: the key negative result

FAILED ON THIS BENCHMARK

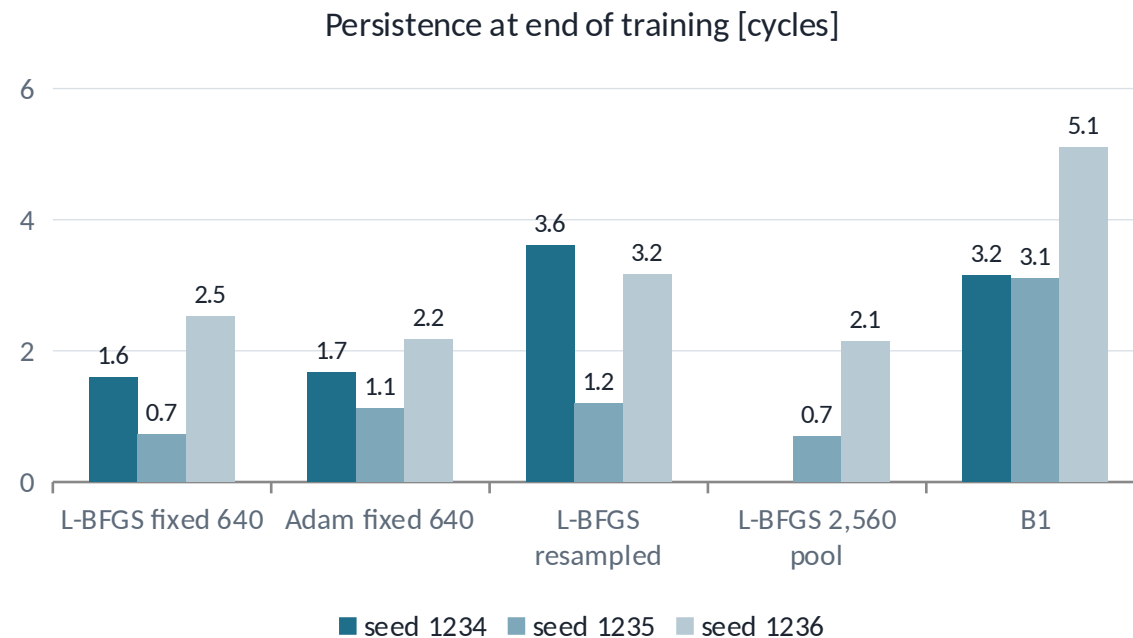
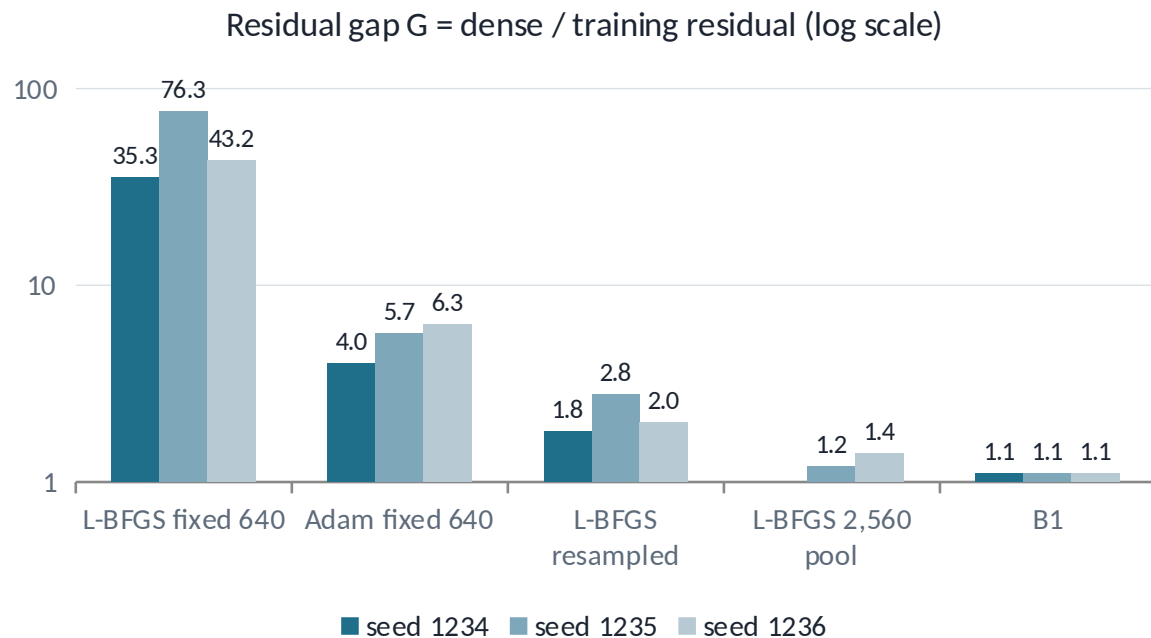


Seed	Persistence change [cycles]	L2 vs B1	PDE residual vs B1
1234	-1.6	×1.28	×1.55
1235	-2.4	×1.54	×2.38
1236	-2.6	×1.44	×1.22

Cheaper in time (0.87–0.89× wall-clock) but 1.61–1.63× peak RAM, and worse on every accuracy metric in 3/3 seeds.

The optimizer that solves the isolated temporal problem does not automatically solve the coupled space–time problem.

B2-E03: sparse collocation alone is not the root cause



Residual gap G: L-BFGS fixed 640 35–76× · Adam fixed 640 4–6× · L-BFGS resampled 1.8–2.8× · L-BFGS 2,560 pool 1.2–1.4× · B1 1.1×

L-BFGS 2,560 pool, seed 1234: diverged at 576,640 evaluations (no bar; excluded by pre-registered Amendment 1).

More collocation coverage removed the train/dense residual gap without restoring dynamic propagation. Sparse collocation alone is not the root cause — no stage-2 variant beat B1 (pre-registered CASE D).

Where we are: no optimized full-field PINN yet

- | | | |
|---|--|--------------|
| 1 | Benchmark established | ACHIEVED |
| 2 | Reproducibility established | ACHIEVED |
| 3 | Failure mechanisms narrowed (refinement ongoing) | ACHIEVED |
| 4 | Candidate optimization mechanisms identified | PARTIAL |
| 5 | Reproducible full-field improvement | NOT YET DONE |
| 6 | Compute-efficient optimized PINN | NOT YET DONE |
| 7 | Railway validation | NOT YET DONE |

Batch-1 best full-field run: L2 0.263, 8.09 cycles at 4× compute (2.56M evaluations, 1 seed), still collapsing at 0.393 s.

Full-field reference (B1, 640k evaluations)

L2 0.526
3.15 cycles

seed 1234; across 3 seeds L2 0.402–0.549, 3.11–5.10 of 20.58 cycles

Best modal diagnostic (Adam → L-BFGS)

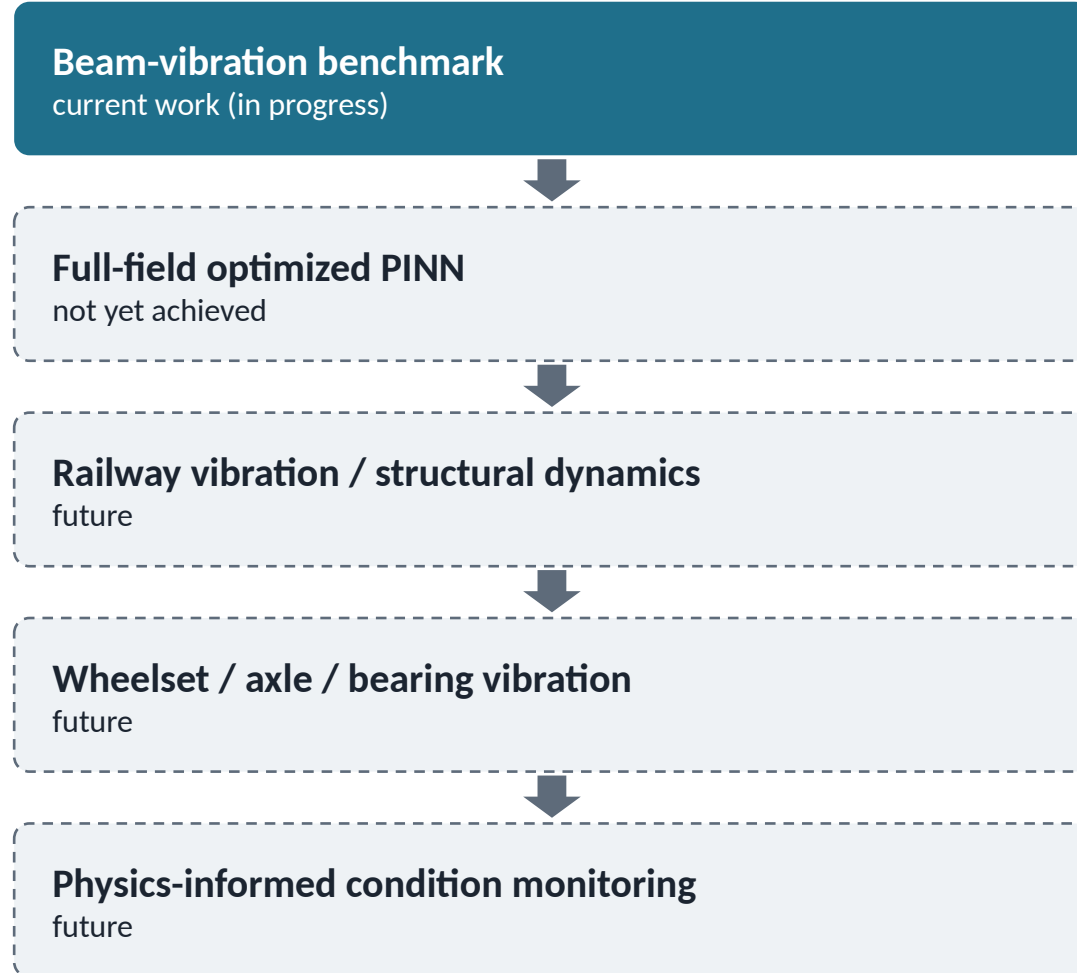
L2 0.038–0.093
20.58 cycles

3/3 seeds • exact spatial mode supplied — not the optimized PINN

Closing this gap on the coupled full field is the central open problem.

Path to the railway application

FUTURE WORK



Why start from a beam benchmark

- Exact solution available, so dynamic fidelity (frequency, damping, persistence) is measurable
- Euler-Bernoulli beam models are standard building blocks of track and bridge dynamics in the PINN literature (e.g. beams on Winkler foundations, moving loads)
- A method must first preserve dynamics on controlled physics before it is trusted on noisy railway measurements

No railway experiment, dataset or deployment has been performed yet. Railway work starts only after a validated full-field method exists.

Takeaways

- 1 We have not found the final optimized PINN yet.
- 2 We have ruled out several obvious explanations — precision, derivative order alone, collocation density, the L-BFGS switch alone — and identified a strong gap between temporal-only and coupled space-time learning.
- 3 The next breakthrough must improve the full-field solution, not merely the diagnostic modal problem.

Research objective

Lower error

+

Longer physical fidelity

+

Less compute

+

Transfer to railway physics